

Halliday & Resnick's

Fundamentals of Physics

By Jearl Walker

Chapter 28

Magnetic Fields

28.2: What Produces Magnetic Field?:

1) One way that magnetic fields are produced is to **use moving electrically charged particles, such as a current in a wire, to make an electromagnet**. The current produces a magnetic field that is utilizable.

2) The other way to produce a magnetic field is by means of **elementary particles such as electrons**, because these particles have an ***intrinsic magnetic field*** around them.

3) The magnetic fields of the electrons in certain materials add together to give a net magnetic field around the material. Such addition is the reason why a **permanent magnet**, has a permanent magnetic field.

In other materials, the magnetic fields of the electrons cancel out, giving no net magnetic field surrounding the material.

28.3: The Definition of $\mathbf{B}$:

We can define a **magnetic field, $\mathbf{B}$** , by firing a charged particle through the point at which is to be defined, using various directions and speeds for the particle and determining the force that acts on the particle at that point.

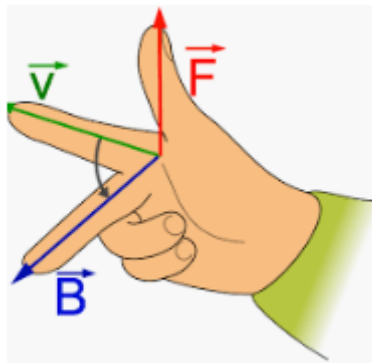
The magnetic force on the charged particle, $\mathbf{F}_B$, is defined to be:

$$\vec{F}_B = q\vec{v} \times \vec{B}$$

Here q is the charge of the particle, $\mathbf{v}$ is its velocity, and $\mathbf{B}$ the magnetic field in the region. The magnitude of this force is then:

Here ϕ is the angle between vectors $\mathbf{v}$ and $\mathbf{B}$.

$$F_B = |q|vB \sin \phi,$$



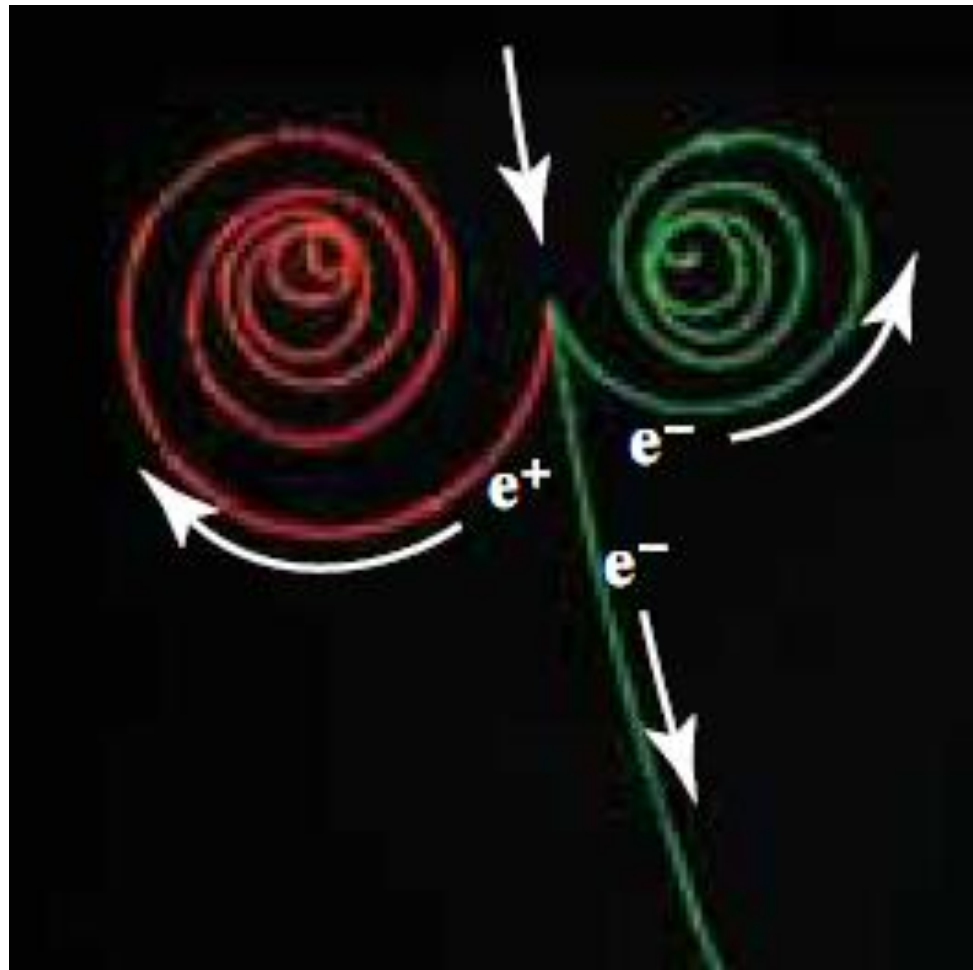


Fig. 28-3 The tracks of two electrons and a positron in a bubble chamber that is immersed in a uniform magnetic field that is directed out of the plane of the page.

(Lawrence Berkeley Laboratory/Photo Researchers)

28.3: Finding the Magnetic Force on a Particle:

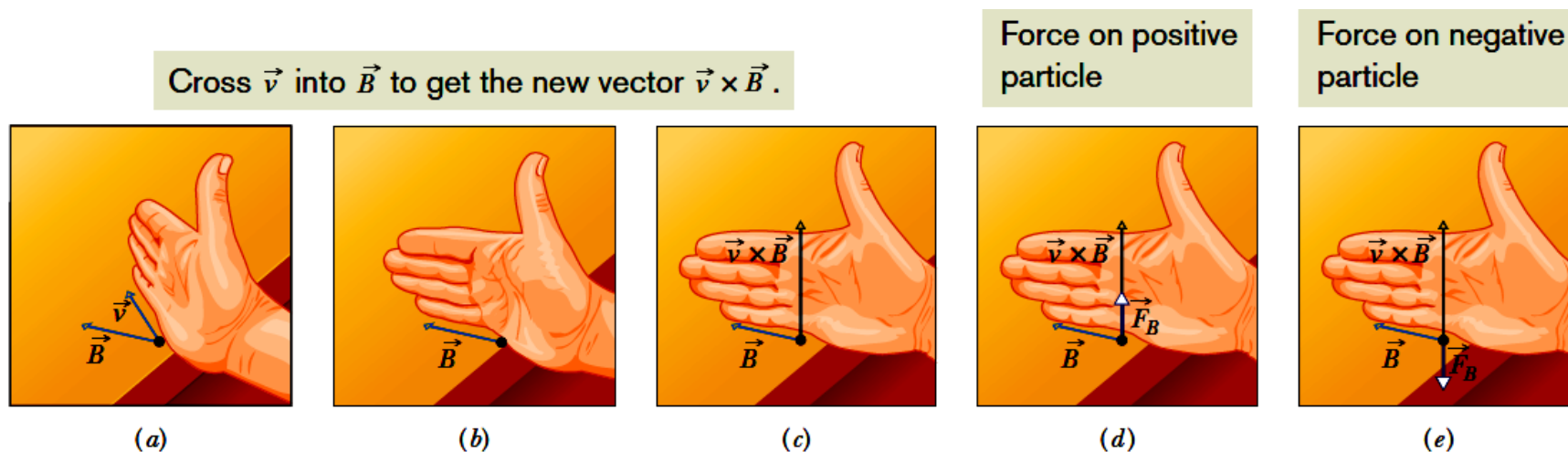


Fig. 28-2 (a) – (c) The right-hand rule (in which $\vec{v}$ is swept into $\vec{B}$ through the smaller angle ϕ between them) gives the direction of $\vec{v} \times \vec{B}$ as the direction of the thumb. (d) If q is positive, then the direction of $\vec{F}_B = q\vec{v} \times \vec{B}$ is in the direction of $\vec{v} \times \vec{B}$. (e) If q is negative, then the direction of $\vec{F}_B$ is opposite that of $\vec{v} \times \vec{B}$.



The force $\vec{F}_B$ acting on a charged particle moving with velocity $\vec{v}$ through a magnetic field $\vec{B}$ is *always* perpendicular to $\vec{v}$ and $\vec{B}$.

28.3: The Definition of B :

The SI unit for $\mathbf{B}$ that follows is newton per coulomb-meter per second. For convenience, this is called the **tesla (T)**:

Table 28-1

Some Approximate Magnetic Fields

At surface of neutron star	10^8 T
Near big electromagnet	1.5 T
Near small bar magnet	10^{-2} T
At Earth's surface	10^{-4} T
In interstellar space	10^{-10} T
Smallest value in magnetically shielded room	10^{-14} T

$$1 \text{ tesla} = 1 \text{ T} = 1 \frac{\text{newton}}{(\text{coulomb})(\text{meter/second})}$$
$$= 1 \frac{\text{newton}}{(\text{coulomb/second})(\text{meter})} = 1 \frac{\text{N}}{\text{A} \cdot \text{m}}$$

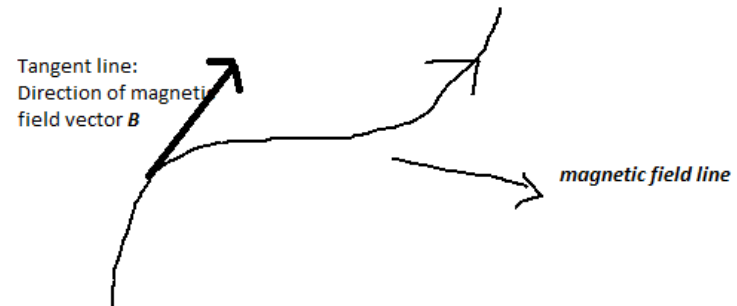
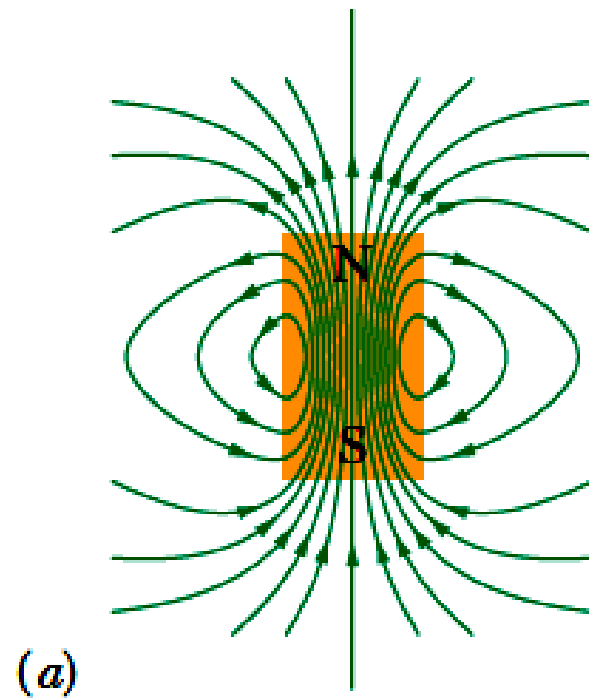
An earlier (non-SI) unit for $\mathbf{B}$ is the *gauss* (G), and

$$1 \text{ tesla} = 10^4 \text{ gauss.}$$

28.3: Magnetic Field Lines:

□ The direction of the tangent to a magnetic field line at any point gives the direction of $\mathbf{B}$ at that point.

□ The spacing of the lines represents the magnitude of $\mathbf{B}$ —the magnetic field is stronger where the lines are closer together, and conversely.



Opposite magnetic poles attract each other, and like magnetic poles repel each other.

28.6: A Circulating Charged Particle:

Consider a particle of charge magnitude $|q|$ and mass m moving perpendicular to a uniform magnetic field $\mathbf{B}$, at speed v .

The magnetic force continuously deflects the particle, and since $\mathbf{B}$ and $\mathbf{v}$ are always perpendicular to each other, this deflection causes the particle to follow a circular path.

The magnetic force acting on the particle has a magnitude of $|q|vB$. For uniform circular motion $F = m \frac{v^2}{r}$,

$$\Rightarrow |q|vB = \frac{mv^2}{r} \quad \text{magnetic force} = \text{centripetal force}$$

$$F_b = F_c$$

$$s = vT$$

circumference of the circle formed by the charged particle

$$T = s/v$$

$$s = 2\pi r$$

$$r = \frac{mv}{|q|B} \quad (\text{radius}).$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \frac{mv}{|q|B} = \frac{2\pi m}{|q|B} \quad (\text{period}).$$

$$f = \frac{1}{T} = \frac{|q|B}{2\pi m} \quad (\text{frequency}).$$

$$\omega = 2\pi f = \frac{|q|B}{m} \quad (\text{angular frequency}).$$

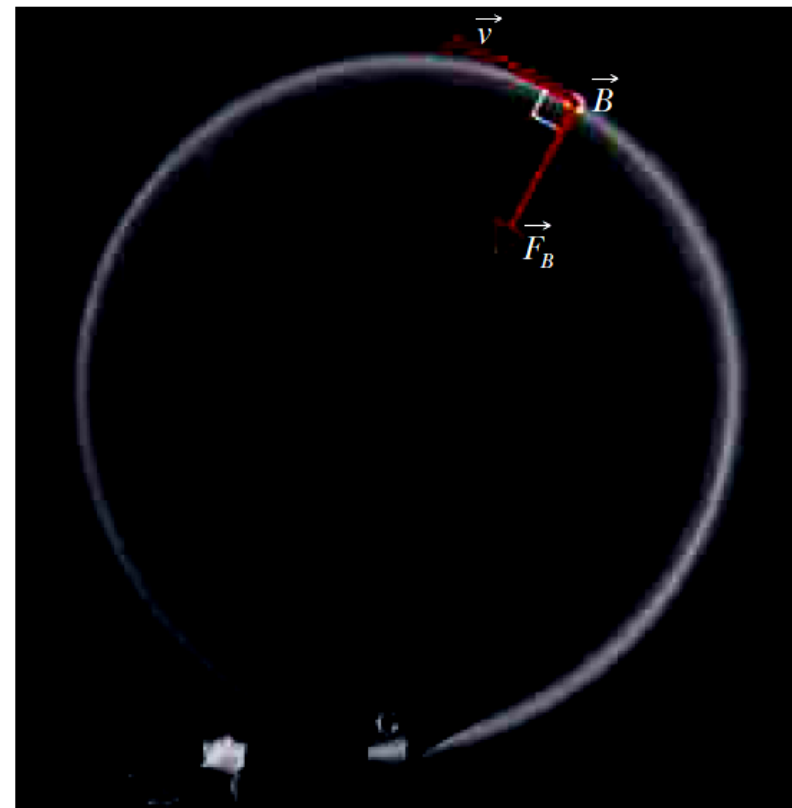


Fig. 28-10 Electrons circulating in a chamber containing gas at low pressure (their path is the glowing circle).

A uniform magnetic field, B , pointing directly out of the plane of the page, fills the chamber.

Note the radially directed magnetic force F_B ; for circular motion to occur, F_B must point toward the center of the circle,

(Courtesy John Le P. Webb, Sussex University, England)

Example, Magnetic Force on a Moving Charged Particle :

A uniform magnetic field $\vec{B}$, with magnitude 1.2 mT, is directed vertically upward throughout the volume of a laboratory chamber. A proton with kinetic energy 5.3 MeV enters the chamber, moving horizontally from south to north. What magnetic deflecting force acts on the proton as it enters the chamber? The proton mass is 1.67×10^{-27} kg. (Neglect Earth's magnetic field.)

KEY IDEAS

Because the proton is charged and moving through a magnetic field, a magnetic force $\vec{F}_B$ can act on it. Because the initial direction of the proton's velocity is not along a magnetic field line, $\vec{F}_B$ is not simply zero.

Magnitude: To find the magnitude of $\vec{F}_B$, we can use Eq. 28-3 ($F_B = |q|vB \sin \phi$) provided we first find the proton's speed v . We can find v from the given kinetic energy because $K = \frac{1}{2}mv^2$. Solving for v , we obtain

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{(2)(5.3 \text{ MeV})(1.60 \times 10^{-13} \text{ J/MeV})}{1.67 \times 10^{-27} \text{ kg}}} \\ = 3.2 \times 10^7 \text{ m/s.}$$

Equation 28-3 then yields

$$F_B = |q|vB \sin \phi \\ = (1.60 \times 10^{-19} \text{ C})(3.2 \times 10^7 \text{ m/s}) \\ \times (1.2 \times 10^{-3} \text{ T})(\sin 90^\circ) \\ = 6.1 \times 10^{-15} \text{ N.} \quad (\text{Answer})$$

This may seem like a small force, but it acts on a particle of small mass, producing a large acceleration; namely,

$$a = \frac{F_B}{m} = \frac{6.1 \times 10^{-15} \text{ N}}{1.67 \times 10^{-27} \text{ kg}} = 3.7 \times 10^{12} \text{ m/s}^2.$$

Direction: To find the direction of $\vec{F}_B$, we use the fact that $\vec{F}_B$ has the direction of the cross product $q\vec{v} \times \vec{B}$. Because the charge q is positive, $\vec{F}_B$ must have the same direction as $\vec{v} \times \vec{B}$, which can be determined with the right-hand rule for cross products (as in Fig. 28-2d). We know that $\vec{v}$ is directed horizontally from south to north and $\vec{B}$ is directed vertically up. The right-hand rule shows us that the deflecting force $\vec{F}_B$ must be directed horizontally from west to east, as Fig. 28-6 shows. (The array of dots in the figure represents a magnetic field directed out of the plane of the figure. An array of Xs would have represented a magnetic field directed into that plane.)

If the charge of the particle were negative, the magnetic deflecting force would be directed in the opposite direction—that is, horizontally from east to west. This is predicted automatically by Eq. 28-2 if we substitute a negative value for q .

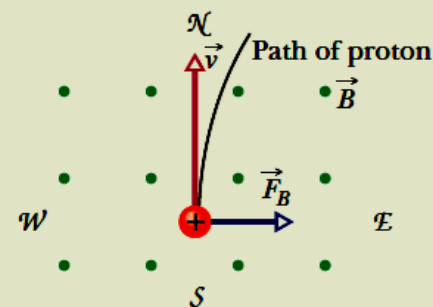


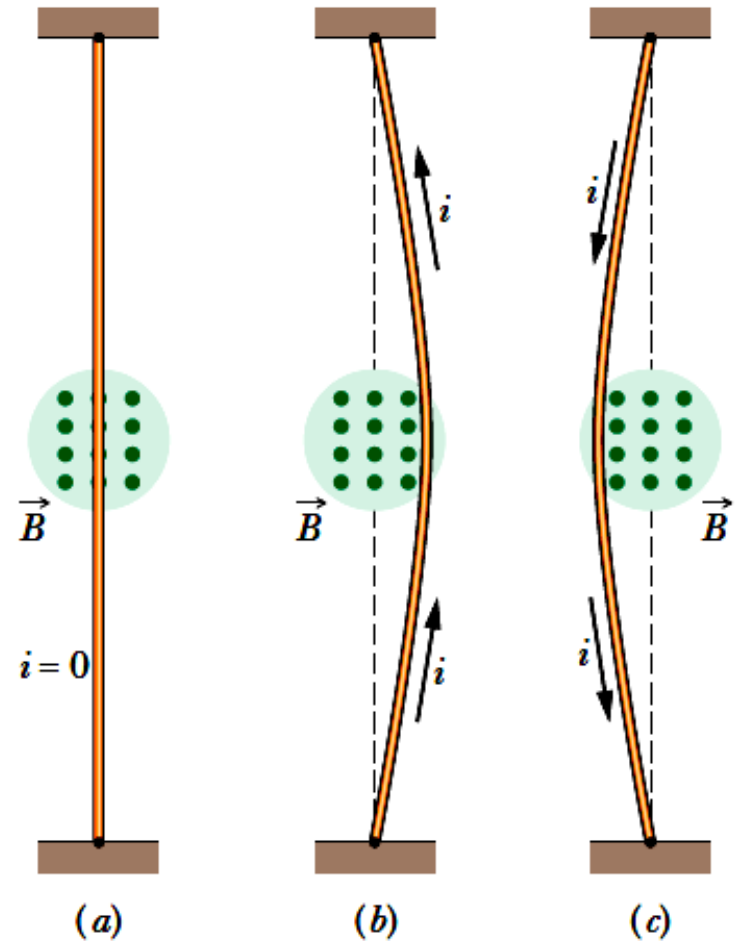
Fig. 28-6 An overhead view of a proton moving from south to north with velocity $\vec{v}$ in a chamber. A magnetic field is directed vertically upward in the chamber, as represented by the array of dots (which resemble the tips of arrows). The proton is deflected toward the east.

28.8: Magnetic Force on a Current-Carrying Wire:

Fig. 28-14 A flexible wire passes between the pole faces of a magnet (only the farther pole face is shown). (a) Without current in the wire, the wire is straight. (b) With upward current, the wire is deflected rightward. (c) With downward current, the deflection is leftward. The connections for getting the current into the wire at one end and out of it at the other end are not shown.

B: out of the page

A force acts on a current through a B field.



28.8: Magnetic Force on a Current-Carrying Wire:

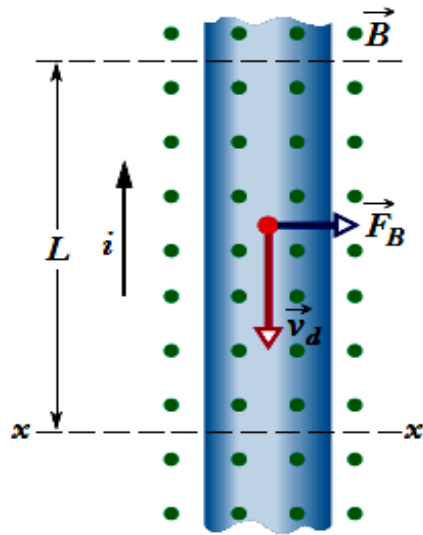


Fig. 28-15 A close-up view of a section of the wire of Fig. 28-14*b*. The current direction is upward, which means that electrons drift downward. A magnetic field that emerges from the plane of the page causes the electrons and the wire to be deflected to the right.

Consider a length L of the wire in the figure. All the conduction electrons in this section of wire will drift past plane xx in a time $t = L/v_d$ (from $s = vt$ formula; where $s = L$ and $v = v_d$)

Thus, in that time a charge will pass through that plane that is given by

$$q = it = i \frac{L}{v_d}$$

$$F_B = qv_d B \sin \phi = \frac{iL}{v_d} v_d B \sin 90^\circ$$

$$F_B = iLB.$$

$$\vec{F}_B = i\vec{L} \times \vec{B} \quad (\text{force on a current}).$$

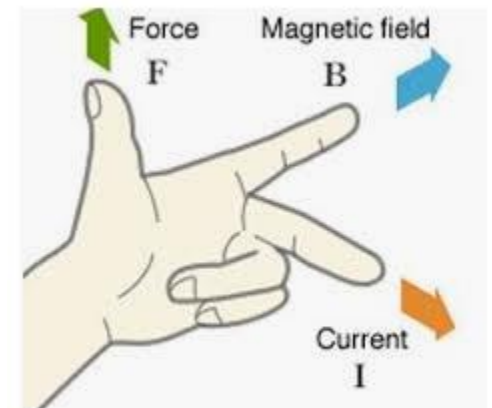
Here $\vec{L}$ is a length vector that has magnitude L and is directed along the wire segment in the direction of the (conventional) current.

$$d\vec{F}_B = i d\vec{L} \times \vec{B},$$

If a wire is not straight or the field is not uniform, we can imagine the wire broken up into small straight segments . The force on the wire as a whole is then the vector sum of all the forces on the segments that make it up.

In the differential limit, we can write and we can find the resultant force on any given arrangement of currents by integrating Eq. 28-28 over that arrangement.

$$d\vec{F}_B = i d\vec{L} \times \vec{B},$$



28.5: Crossed Fields, The Hall Effect:

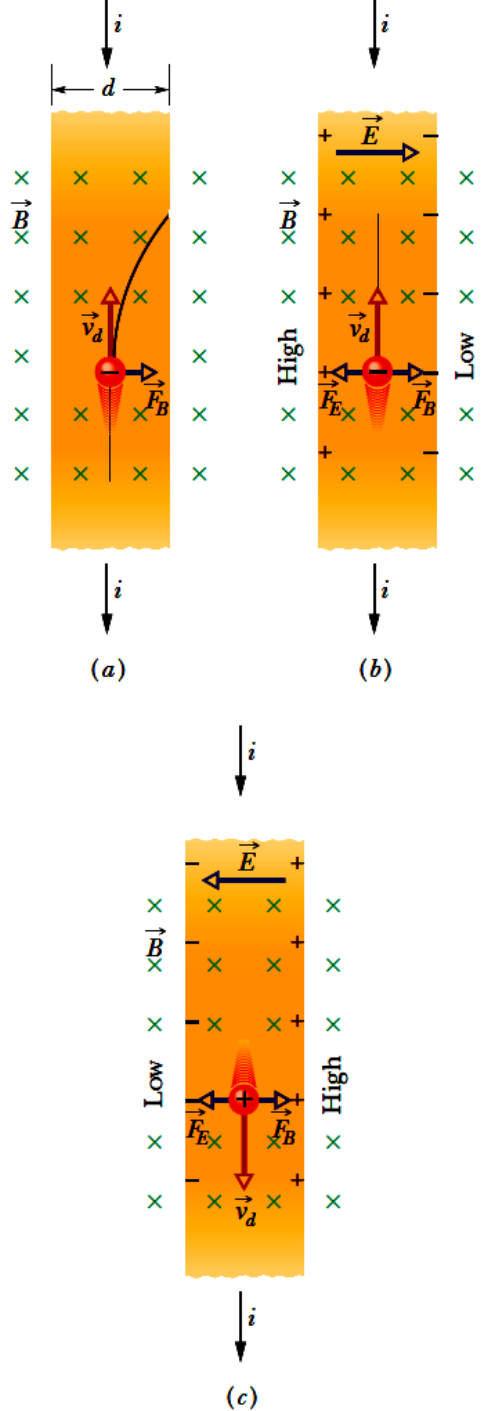
Fig. 28-8 A strip of copper carrying a **current i is immersed in a magnetic field** .

(a) The situation immediately after the magnetic field is turned on. The **curved path** that will then be taken by an electron is shown.

(b) The situation at equilibrium, which quickly follows. Note that **negative charges pile up on the right side** of the strip, leaving **uncompensated positive charges on the left**. Thus, the left side is at a higher potential than the right side.

(c) For the same current direction, if the charge carriers were positively charged, *they would pile up on the right side*, and the right side would be at the higher potential.

B: into the page



A Hall potential difference V is associated with the electric field across strip width d , and the magnitude of that potential difference is $V = Ed$.

When the electric and magnetic forces are in balance (Fig. 28-8b),

where v_d is the drift speed. But,

$$eE = ev_d B.$$

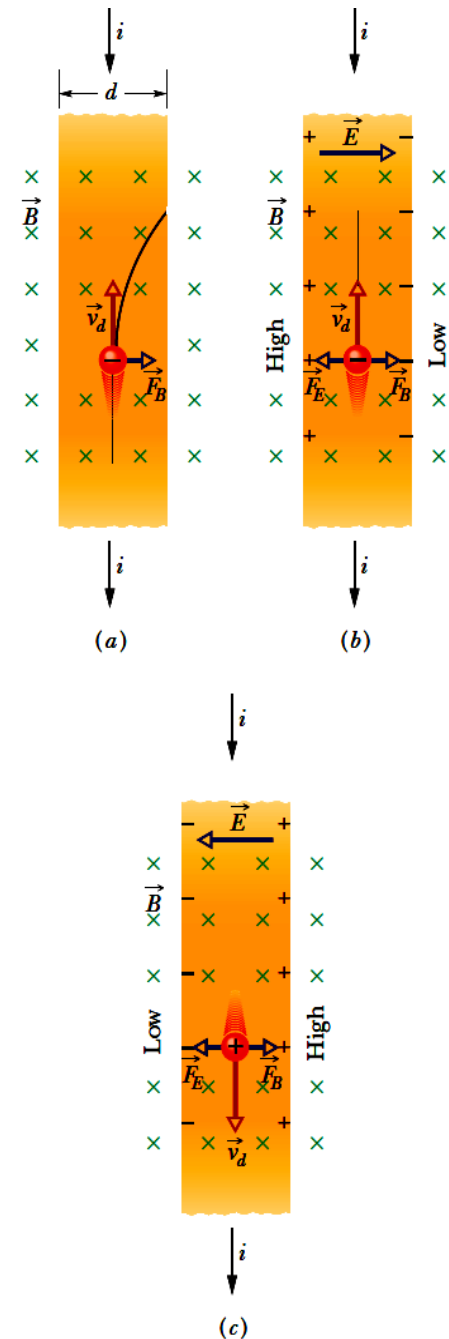
$$v_d = \frac{J}{ne} = \frac{i}{neA}$$

Where J is the current density, A the cross-sectional area, e the electronic charge, and n the number of charges per unit volume.

Therefore,

$$n = \frac{Bi}{Vle},$$

Here, $l = (A/d)$, the thickness of the strip.



HALL EFFECT

$$F_E = F_B$$

$$eE = ev_d B$$

$$E = v_d B \quad - (1)$$

$$J = nev_d$$

$$\frac{J}{ne} = v_d$$

$$J = \frac{i}{A}$$

$$v_d = \frac{i}{Ane}$$

$$A = lxd$$

$$v_d = \frac{i}{ldne} \quad - (2)$$

$$V = Ed. \Rightarrow E = \frac{V}{d} \quad - (3)$$

(P.D b/w the left & the right side of the strip)

$$E = v_d B$$

$$\frac{V}{d} = \frac{i}{Ane} B$$

$$V = \frac{iB}{lne}$$

$$\frac{B i}{V l e} = n$$

$l \rightarrow$ thickness } of the
 $d \rightarrow$ width } strip.

Example, Magnetic Force on a Wire Carrying Current:

A straight, horizontal length of copper wire has a current $i = 28 \text{ A}$ through it. What are the magnitude and direction of the minimum magnetic field $\vec{B}$ needed to suspend the wire—that is, to balance the gravitational force on it? The linear density (mass per unit length) of the wire is 46.6 g/m .

KEY IDEAS

(1) Because the wire carries a current, a magnetic force $\vec{F}_B$ can act on the wire if we place it in a magnetic field $\vec{B}$. To balance the downward gravitational force $\vec{F}_g$ on the wire, we want $\vec{F}_B$ to be directed upward (Fig. 28-17). (2) The direction of $\vec{F}_B$ is related to the directions of $\vec{B}$ and the wire's length vector $\vec{L}$ by Eq. 28-26 ($\vec{F}_B = i\vec{L} \times \vec{B}$).

Calculations: Because $\vec{L}$ is directed horizontally (and the current is taken to be positive), Eq. 28-26 and the right-hand rule for cross products tell us that $\vec{B}$ must be horizontal and rightward (in Fig. 28-17) to give the required upward $\vec{F}_B$.

The magnitude of $\vec{F}_B$ is $F_B = iLB \sin \phi$ (Eq. 28-27). Because we want $\vec{F}_B$ to balance $\vec{F}_g$, we want

$$iLB \sin \phi = mg, \quad (28-29)$$

where mg is the magnitude of $\vec{F}_g$ and m is the mass of the wire.

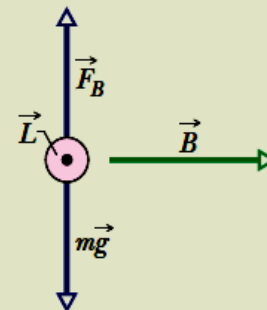


Fig. 28-17 A wire (shown in cross section) carrying current out of the page.

We also want the minimal field magnitude B for $\vec{F}_B$ to balance $\vec{F}_g$. Thus, we need to maximize $\sin \phi$ in Eq. 28-29. To do so, we set $\phi = 90^\circ$, thereby arranging for $\vec{B}$ to be perpendicular to the wire. We then have $\sin \phi = 1$, so Eq. 28-29 yields

$$B = \frac{mg}{iL \sin \phi} = \frac{(m/L)g}{i}. \quad (28-30)$$

We write the result this way because we know m/L , the linear density of the wire. Substituting known data then gives us

$$\begin{aligned} B &= \frac{(46.6 \times 10^{-3} \text{ kg/m})(9.8 \text{ m/s}^2)}{28 \text{ A}} \\ &= 1.6 \times 10^{-2} \text{ T}. \end{aligned} \quad (\text{Answer})$$

This is about 160 times the strength of Earth's magnetic field.